ALESSANDRO CAROTENUTO

MSc Student, AI & Robotics | Computer and Automation Engineer

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AI & Robotics Engineer

MSc student in **AI & Robotics (Sapienza)**, with a BSc in Computer and Automation Engineering. I build models and systems from scratch, from the math and the PyTorch code to a working, tested prototype. I build **robotics, deep learning, reinforcement learning, control systems and computer vision** systems with a strong software foundation. I consider myself both well-versed in theory and research, while keeping a pragmatic view to develop my projects.

WORK EXPERIENCE

AI & Robotics Engineer (Part Time) **Oct 2026 - Present**
SASA: Sapienza Technology Team, Rome, Italy

- Part of a multidisciplinary student team, currently working on the **Kelvis "Lunar X-View"** Computer Vision challenge and on the re-design of the **CC8 Rover Robotic Arm**

Private Tutor (alongside high school, BSc and MSc studies) **2017 - 2026**
Self-Employed, Tivoli, Italy

- Private tutoring in Mathematics, Physics, Calculus and Computer Science for high school and university students

EDUCATION

Master's Degree in Artificial Intelligence & Robotics **2023 - Present**
Sapienza, University of Rome
Coursework: Robot Kinematics & Dynamics, Machine Learning, Reinforcement Learning, Control & System Theory, Computer Vision, Artificial Intelligence, Symbolic AI, Deep Learning, Generative AI

Bachelor's Degree in Computer and Automation Engineering **2018 - 2023**
Sapienza, University of Rome
Coursework: Software Design & Development, Systems & Control Theory, Numerical Analysis, Calculus, Electronics, Telecommunications, System Modelling and Simulation, Operations Research, Probability and Statistics

High School Diploma for Applied Scientific Studies **2013 - 2018**
Lazzaro Spallanzani, Scientific High School

SELECTED PROJECTS

Autonomous Quadrotor Vision-Based Landing
2025 - 2026 *implementation of Falanga et al. 2017*

- Built a vision-based landing pipeline (**MATLAB/Simulink, CoppeliaSim**) with Kalman filtering and **PID control**, achieving ∼**100% simulated success** and **less than 0.5 m** tracking error

Hierarchical RL with World Graph for Navigation
2025 *extension of Shang et al. 2019*

- Built a **hierarchical Reinforcement Learning framework** (PyTorch) for navigation, combining a **VAE-based generative** module to discover pivotal states with graph-based planning, achieving an **86.8% success** rate on multi-goal tasks.

Collision & Connectivity for Robot Swarms via CBF
2025 - 2026

- Designed, in a team, a distributed safety framework (**MATLAB/Simulink**) using **Control Barrier Functions** and consensus-based connectivity estimation, validated **on a 5-robot swarm across 42 obstacles**

Few-Shot Hyperspectral Image classifier w/CTA-Net
2026 *extension of Fu et al. 2025*

- Few-shot HSI classification using a CNN-Transformer architecture with channel-spatial attention and sample amplification, reaching **88.42% OA, 90.07% AA, and 84.96% Kappa** with **around 30 min** training time. Explored adaptive spectral pruning masks.

ROS2 Inverse Kinematics Library for Manipulators
2026

- Manipulator kinematics library **from scratch** that supports both **DH-based** forward kinematics and **URDF** import, a numerical Jacobian, and iterative inverse position kinematics using an SVD pseudoinverse.

Ground2Aerial Generation via Transformers
2025

- Built a model that from ground-level images **generates** plausible satellite views using a two-stage **VQGAN-Transformer** pipeline (PyTorch, Hugging Face, OpenCV), with experiments tracked in WandB.

CERTIFICATIONS, VOLUNTEERING, COMPETITIONS

Cisco IT Essentials Certification *Cisco Systems Inc.* *2018*	**Arduino Workshop for Children** *Maker Faire Rome* *2018*	**NASA Space Apps Challenge** *NASA Space Apps* *2019*

Sapienza Generative AI Hackathon (2026): #3 out of 18 teams: CVAE-based music style transfer

TECHNICAL SKILLS

Core Programming	Python, C++, MATLAB, TypeScript, JavaScript, Java, Simulink, LaTeX
Libraries & Tools	PyTorch, NumPy, ROS2, Linux, scikit-learn, Matplotlib, SimPy, Eigen, Git, WandB, CoppeliaSim
Languages	Italian (native), English (B2/C1)

I hereby authorize the processing of my personal data contained in this CV pursuant to Legislative Decree no. 196/2003 and Art. 13 of GDPR (EU) 679/2016.